CentraleSupelec **CFD**

Computational study of the flow in a lid-driven cavity

Context

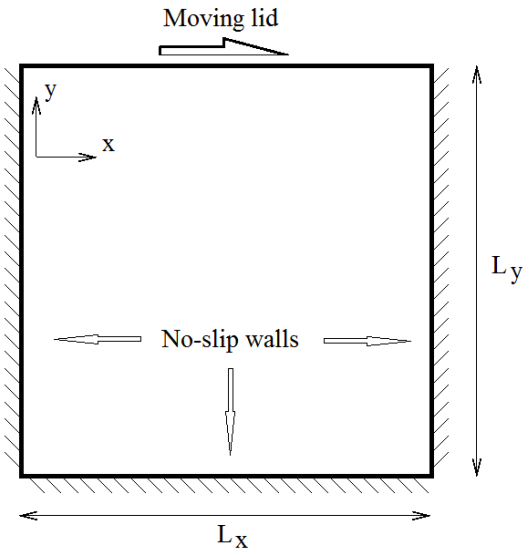
We consider a two-dimensional square cavity of size $L \times L$ filled with a newtonian single-phase fluid. All walls are fixed (no-slip) except the upper lid, which moves tangentially at velocity $U_t(t)$. The flow is governed by the incompressible Navier–Stokes equations only (no scalar transport is considered):

$$\frac{\partial u_i}{\partial x_i} = 0, \qquad \frac{\partial u_j}{\partial t} + \frac{\partial u_i u_j}{\partial x_i} = -\frac{1}{\rho}\frac{\partial P}{\partial x_j} + \nu \frac{\partial^2 u_j}{\partial x_i \partial x_i} \tag{1}$$

The lid is started with the ramp

$$U_t(t) = U\left(1 - \exp\left(-\frac{t}{\tau_{lid}}\right)\right), \qquad \tau_{lid} = \frac{L}{U}, \tag{2}$$

and the regime is controlled by the Reynolds number $Re = UL/\nu$.



The objective of this project is to study the flow regimes in this configuration. In this optic you will develop your own solver(s) and validate them in the most convincing way.

Global diagnostics. Throughout the project, you will monitor (per unit depth):

- the kinetic energy $K(t) = \dfrac{\rho}{2}\displaystyle\int_\Omega (u^2 + v^2)\,\mathrm{d}S$,
- the power injected by the lid $P(t) = \displaystyle\int_0^L \rho\nu \left.\frac{\partial u}{\partial y}\right|_{y=L} U_t(t)\,\mathrm{d}x$,

You are totally free to monitor other quantities.

You have to keep in mind that this is a "realistic" and relatively open study: you can end up failing into simulating specific aspects of the flow because of the limitation of your methods. Therefore it is up to you

to explain your choices, your study, your diagnostic, to convince us of the rationale behind the possible failure in specific regimes.

Choice of methods. A key objective of the project is to compare differents numerical methods, to see what could be the impact of a wise or unwise choice of methods. Therefore, we require at least two methods to be tested. We require at least one method based on the content of this course, but you are free to use more advance methods (but you need to understand them).

Part A – A verified reference solver at low Reynolds number

- Compute the steady flow at $Re = 100$.
- Validate your results either with results from the literature and/or by comparing with the results of another validated solver. Justify that your grid is sufficient.

Part B – Building a solver for high Reynolds numbers

- Compute the flow for differents Reynols number with $Re < 7500$. Describe the evolution of the flow topology and compare with the literature and/or with the results of another validated solver.
- Plot P as a function of Re and identify a scaling law.

Part C – High reynolds numbers and loss of stationarity

Above a critical Reynolds number Re_c (the literature reports values close to 8000 for the square cavity), the steady solution becomes unstable and the flow becomes periodic.

- Simulate the flow at the highest Reynolds number you can afford, to be justified.
- Investigate the sensitivity of the results of this highest Reynolds number to the numerical choices.
- Compare you results with the literature and/or with the results of another validated solver.